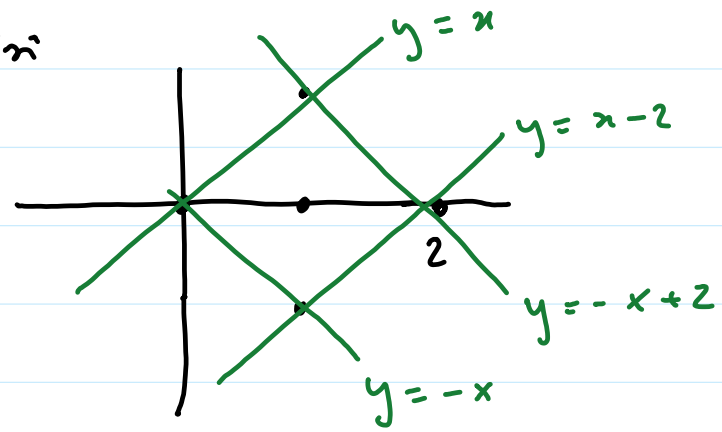
Appello 3

martedì 19 luglio 2016 11:37

Analisi

3.



$$D = \{(x,y): \quad x-y \in [0,2], \quad x+y \in [0,2]\}$$

Posto $u = x-y$, $v = x+y$ è $\begin{cases} x = \frac{u+v}{2} \\ y = \frac{v-u}{2} \end{cases} = \phi(u,v)$

$$\text{Jac } \phi(u,v) = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2}$$

$$\frac{x-y}{x+y+1} = \frac{u}{v+1} \quad \text{e}$$

$$\int_D \sqrt{\frac{x-y}{x+y+1}}\, dx\, dy = \int_{[0,2]\times[0,2]} \frac{1}{2}\sqrt{\frac{u}{v+1}}\, du\, dv$$

$$= \frac{1}{2} \int_0^2 \sqrt{u}\, du \int_0^2 \frac{1}{\sqrt{v+1}}\, dv$$

$$= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}}\right]_0^2 \left[2(v+1)^{\frac{1}{2}}\right]_0^2 = \frac{1}{3} \cdot 2^{3/2} \cdot 2(\sqrt{3}-1)$$

$$= \frac{4}{3}\sqrt{2}(\sqrt{3}-1)$$

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$$\int_\gamma \nabla(1 + x^2 y z) \cdot d\gamma = \left[1 + x^2 y z\right]_{\gamma(0)}^{\gamma(\pi)} = 0$$

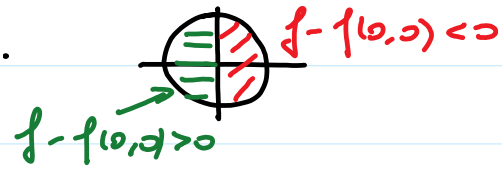
4. $\displaystyle\lim_{(0,0)} \frac{y^4 - 2x^2}{y^4 + x^2}$ non esiste. Infatti

su $x = m y^2$ è $\displaystyle\frac{y^4 - 2x^2}{y^4 + x^2} = \frac{1 - 2m^2}{1 + m^2}$.

2. $f(x,y) = 2 - xy^2$. $\quad \partial_x f(x,y) = -y^2 \quad \partial_y f(x,y) = -2xy$

$$\mathrm{Hess} f(x,y) = \begin{pmatrix} 0 & -2y \\ -2y & -2x \end{pmatrix} \Rightarrow \mathrm{Hess} f(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

L'Hessiano non permette di concludere.
Si ha $f(x,y) - f(0,0) = (2 - xy^2) - 2 = -xy^2$ ed è $xy^2 < 0$ se $x < 0$, $xy^2 > 0$ se $x > 0$: in ogni intorno di $(0,0)$ $f(x,y) - f(0,0)$ assume valori di ambo i segni e quindi $(0,0)$ è di SELLA.



5. $\begin{cases} y' = y/10 - 40 \\ y(0) = 500 \end{cases}$

$$y' = y/10 - 40 \Rightarrow y' - \frac{y}{10} = -40$$

$$\Rightarrow \left(y e^{-t/10}\right)' = -40 e^{-t/10}$$

$$\Rightarrow y(t) e^{-t/10} - y(0) = -40 \int_0^t e^{-s/10}\, ds$$

$$\Rightarrow y(t)e^{-t/10} - y(0) = -40\int_0 e^{-s/10}\,ds$$
$$= -40\left(-10e^{-t/10}+10\right)$$
$$= 400e^{-t/10} - 400$$

$$\Rightarrow y(t) = 500e^{t/10} + 400 - 400e^{t/10}$$
$$= 100e^{t/10} + 400.$$

Si ha $y(t) = 600 \iff 100e^{t/10} = 200$
$$\iff e^{t/10} = 2 \iff t/10 = \lg 2$$
$$\iff t = 10\lg 2.$$

Probabilità

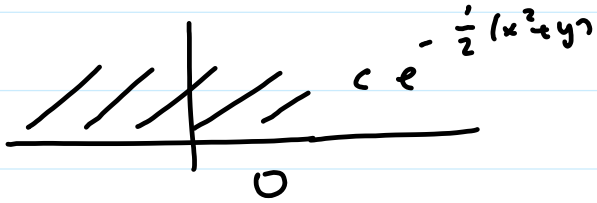
1. $P(\text{Rosse} \mid \text{Stesso Colore}) = \dfrac{P(\text{Stesso Colore} \mid \text{Rosse})P(\text{Rosse})}{P(\text{Stesso Colore})}$

$$P(\underbrace{\text{Stesso Colore}}_{S}) = P(S \mid R)\underset{\substack{\uparrow\\ 2\text{ Rosse}}}{P(R)} + P(S \mid N)\underset{\substack{\uparrow\\ 2\text{ Nere}}}{P(N)} + P(S \mid V)\underset{\substack{\uparrow\\ 2\text{ Verdi}}}{P(V)}$$
$$= P(R) + P(N) + P(V)$$
$$= \frac{5\times 4 + 3\times 2 + 4\times 3}{12\times 11} = \frac{38}{12\times 11}$$

$$P(R) = \frac{5\times 4}{12\times 11}$$

$$\Rightarrow P(\text{Rosse} \mid \text{St. Colore}) = \frac{5\times 4}{38} = \frac{10}{19}.$$

2.



a) Deve essere $\displaystyle\int_{\mathbb{R}\times[0,+\infty[} c\,e^{-\frac{1}{2}(x^2+y)}\,dx\,dy = 1$

$$\iff c \int_{\mathbb{R}} e^{-x^2/2} \int_0^{+\infty} e^{-y/2}\, dy = 1$$

$$\left.\begin{aligned} &\int_{\mathbb{R}} e^{-x^2/2}\, dx \underset{\left(t=\frac{x}{\sqrt{2}}\right)}{=} \sqrt{2} \int_{\mathbb{R}} e^{-t^2}\, dt = \sqrt{2\pi} \\ &\int_0^{+\infty} e^{-y/2}\, dy = -\left[2e^{-y/2}\right]_0^{\infty} = 2 \end{aligned}\right\} \begin{aligned} &\Rightarrow 2c\sqrt{2\pi} = 1 \\ &\Rightarrow c = \frac{1}{\sqrt{8\pi}} \end{aligned}$$

(b) • $\forall x \quad f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\, dy$

$$= c \int_0^{+\infty} e^{-\frac{1}{2}x^2} e^{-y/2}\, dy = c\, e^{-\frac{1}{2}x^2} \int_0^{+\infty} e^{-y/2}\, dy$$

$$= 2c\, e^{-\frac{1}{2}x^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

• Se $y<0 \quad f_Y(y)=0$

Se $y \geq 0 \quad f_Y(y) = c \int_{\mathbb{R}} e^{-\frac{1}{2}x^2} e^{-y/2}\, dx = c\, e^{-y/2} \int_{\mathbb{R}} e^{-\frac{1}{2}x^2}$

$$= \sqrt{2\pi}\, c\, e^{-y/2}$$

$$= \frac{1}{2} e^{-y/2}.$$

(c) $f_X \neq f_Y$: X e Y non sono ident. distribuite.

Si ha poi $f_X(x) f_Y(y) = \begin{cases} \frac{1}{\sqrt{8\pi}} e^{-\frac{1}{2}x^2 - \frac{1}{2}y} & \text{se } y>0 \\ 0 & \text{altrim.} \end{cases} = f_{X,Y}(x,y)$

$\Rightarrow X$ e Y sono indipendenti.

(d) X, Y indip $\Rightarrow E(XY) = E(X)E(Y) = 0$ dato che $E(X)=0$.

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(e) $P(Y > X^2) = \displaystyle\int_{x^2 < y} f_{X,Y}(x,y)\,dx\,dy$

$$= \int_{\mathbb{R}} \left\{ \int_{x^2}^{+\infty} c\, e^{-\frac{1}{2}x^2 - \frac{1}{2}y}\, dy \right\} dx$$

$$= \int_{\mathbb{R}} c\, e^{-\frac{1}{2}x^2} \left[-2 e^{-\frac{y}{2}} \right]_{y=x^2}^{+\infty} dx = \int_{\mathbb{R}} c\, e^{-\frac{1}{2}x^2} \cdot 2 e^{-\frac{x^2}{2}}\, dx$$

$$= 2c \int_{\mathbb{R}} e^{-x^2}\, dx = 2c\sqrt{\pi} = \frac{1}{\sqrt{2\pi}}\sqrt{\pi} = \frac{1}{\sqrt{2}}.$$

3. $X_1 + \dots + X_n \sim Po(4n) \Rightarrow P(S_n = k) = (4n)^k \dfrac{e^{-4n}}{k!} \quad \forall k$

$Var(S_n) = E(S_n) = 4n$

T. C. Limite $\Rightarrow P(S_n > 390) \sim P(4n + \sqrt{4n}\, Z > 390)$

con $Z \sim N(0,1)$

$$= P\left(Z > \frac{390 - 4n}{2\sqrt{n}}\right) = 1 - \Phi\left(\frac{390 - 4n}{2\sqrt{n}}\right)$$

sol è $P(S_n > 390) > 0.5$

$$1 - \Phi\left(\frac{390 - 4n}{2\sqrt{n}}\right) > 0.5 \Leftrightarrow \Phi\left(\frac{390 - 4n}{2\sqrt{n}}\right) < 0.5 = \Phi(0)$$

$$\Leftrightarrow \frac{390 - 4n}{2\sqrt{n}} < 0 \Leftrightarrow 390 < 4n$$

$$\Leftrightarrow n > \frac{390}{4} = \frac{195}{2}.$$